Automatic Quick-Phase Detection in Bedside Recordings from Patients with Acute Dizziness and Nystagmus

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ABSTRACT

Benign Paroxysmal Positional Vertigo (BPPV) is the most common cause of vertigo. It can be diagnosed and treated with simple maneuvers done by vestibular experts. However, there is a high rate of misdiagnosis that results in high medical costs from unnecessary neuroimaging tests. Here we show how to improve saccade detection methods for automatic detection of quick-phases of nystagmus, a key sign of BPPV. We test our method using eye movement data recorded in patients during the diagnostic maneuver.

CCS CONCEPTS

• **Theory of computation** → *Design and analysis of algorithms*; • **Computing methodologies** → *Unsupervised learning*; • **Applied computing** → *Health care information systems.*

KEYWORDS

Saccade Detection, Benign Vestibulopathies, Vertigo, BPPV

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1 INTRODUCTION

Around 3.3% of the primary complaints from patients presenting to Emergency Departments (ED) are dizziness or vertigo [Newman-Toker et al. 2008]. These cases are frequently misdiagnosed [Kattah Jorge C. et al. 2009] and incur significant medical costs [von Brevern et al. 2006]. To avoid unnecessary testing and patient harm, it is important to distinguish between patients affected by benign peripheral disease and patients suffering from central disorders such as stroke. Automatic analysis of eye movement recordings can provide key information to avoid such diagnostic errors. For example, BPPV, the most common cause of dizziness [Hanley et al. 2001; Katsarkas 1999], can be diagnosed by the presence of a brisk nystagmus (Figure 1) during the Dix-Hallpike maneuver that decays within 30s [Dix and Hallpike 1952].

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2 RESEARCH OBJECTIVES

Here we will focus on automatic quantification of nystagmus recorded at the bedside during the Dix-Hallpike Maneuver with the objective of identifying patients suffering from BPPV. Specifically, we will adapt saccade detection methods to identify quick phases of nystagmus. We will also compare our algorithm with other common saccade detection algorithms using data collected at the bedside from patients suffering from vertigo and dizziness.

Given the similarities between quick phases and saccades, we can rely on previously published saccade detection algorithms, which range from simple threshold algorithms to modern machine learning techniques [Daye and Optican 2014; Engbert and Kliegl 2003; Erkelens and Sloot 1995; Hessels et al. 2017; Hoppe and Bulling 2016; Komogortsev and Karpov 2013; Korda et al. 2018; Mihali et al. 2017; Mould et al. 2012; Mulligan 2018; NystrÃűm and Holmqvist 2010; Otero-Millan et al. 2014; Pander et al. 2014; Santini et al. 2015; Sheynikhovich et al. 2018; Toivanen et al. 2015; Zemblys et al. 2018]. Most of these methods have not been tested in data with nystagmus or with recordings from patients at the bedside, where overall statistics and physiological properties of eye movements may differ from normal recordings. Hence, we plan to expand on these methods for quick-phase detection in presence of nystagmus.

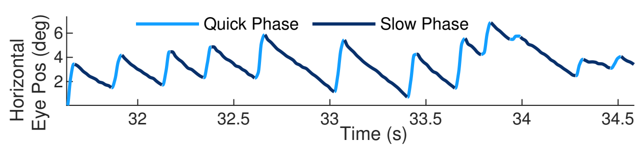

Figure 1: Example of Nystagmus. Pattern of alternating slow phases or eye drift and quick phases or saccades in the opposite direction.

3 HYPOTHESIS AND PROBLEM STATEMENT

To achieve accurate quantification of nystagmus, the method must:

- Be robust to noise and artifacts
- Be able to differentiate between slow and quick phases

4 APPROACH AND METHOD

4.1 Data Resource

We will analyze data collected as part of an ongoing study that compares diagnostic accuracy between eye movement based diagnosis and standard diagnosis. Patients presenting with a chief complaint of acute vertigo were subjected to battery of tests including the Dix-Hallpike positional test. Data was collected using a portable Video Oculography (VOG) device, ICS Impulse, with Otosuite software (GN Otometrics, Taastrup, Denmark) at a rate of 60Hz. Eye position was calibrated using laser targets projected from the goggles.

4.2 Data Preprocessing

To remove artifacts, we first eliminate portions where the horizontal or vertical components present with positions, velocities, or accelerations outside the physiological range expected for humans. The cleaned data was then resampled to 500Hz and filtered with a 2nd order Savitzky Golay filter with window length of 11 samples.

4.3 Quick-Phase (QP) Detection Method

The proposed algorithm first finds all velocity peaks in the data. Then, a subset of peaks is selected aiming to include all true QPs and a similar number of peaks corresponding to noise given the fact that QP rate is known to be bounded [Otero-Millan et al. 2014]. From the selected peaks, different features are extracted, which serve as input for clustering to separate them into QPs and non-QPs.

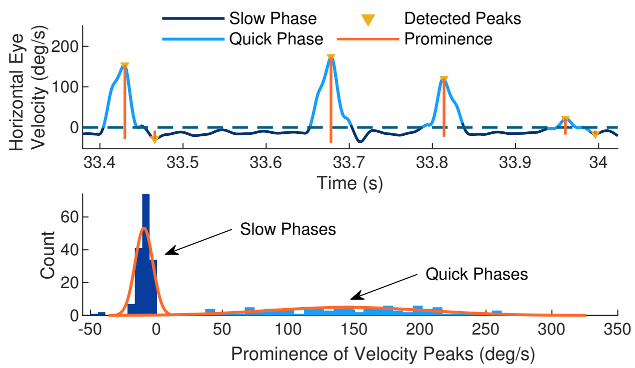

Figure 2: (Top) Eye velocity obtained from data with nystagmus. Note how the second to last peak has a relatively low peak velocity but high prominence. (Bottom) Distribution of prominence of velocity peaks.

4.3.1 Peak Detection. Velocity peaks are detected independently in horizontal and vertical traces. We chose to separate them because vertical data is usually noisier than horizontal data, and nystagmus can occur only in horizontal or vertical components or in both.

4.3.2 Obtaining Data Points for Clustering. An adapted version of the EK method [Engbert and Kliegl 2003] with λ=5 is applied on the prominence of the peaks (measured relative to nearby peaks) rather than on the peak heights (measured from the zero value) because in the presence of nystagmus, there will be a non-zero baseline corresponding to slow-phase velocity as shown in Figure 2. This gives a selection of clear QPs occurring at a rate R1 (peaks/s). We then select, for each one-second time window, a number of peaks equal to $R1 * 2$ with largest prominence so as to include a similar number of true peaks and peaks resulting from noise.

4.3.3 Feature Selection. From the selected peaks, the start and end points for each peak are determined with an angle-based parameter (α) in the velocity space. These periods are then used to calculate different features including amplitude, velocity, acceleration, radius of curvature, median/mean difference before and after detected peak (in position and velocity), and difference in alpha before and after detected peak (in position and velocity). Spearman rank correlation matrix is then used for selecting the important features.

4.3.4 Clustering. A density-based clustering approach [Rodriguez and Laio 2014; Sheynikhovich et al. 2018] was applied on the selected features. This approach involves computing distance between each of the data points. We have applied a cosine distance metric in order to get the distance matrix. By calculating density (ρ_i) of each point via an exponential kernel and its distance (γ_i) to a higher density point, a clear delineation between the cluster centers and rest of the points can be seen. The threshold distance (d_c) to demarcate a cluster is chosen such that the average number of neighbors is less than half of the total number of data points.

5 PRELIMINARY RESULTS

The most important features obtained from the Spearman rank correlation matrix are velocity, median difference before and after peak (in position and velocity), prominence, acceleration, radius of curvature and difference in α between the peak and its previous sample. These features were then used for clustering. Figure 3 shows the results of detected QPs by the present algorithm in presence of nystagmus. Two other methods, EK [Engbert and Kliegl 2003] and NH [NystrÃűm and Holmqvist 2010] were also employed. The NH method was unable to detect most of the quick phases while the EK method (λ = 6) detected most quick phases but also false positives.

6 PLANS FOR FUTURE WORK

The next step in this study includes obtaining a reliable parameter of slow phase velocity (SPV) to quantify nystagmus. From these SPV values, ROC curves will be obtained from the nystagmus data using presence of nystagmus as marker to quantify the accuracy of the proposed method. Additionally, there is a need to investigate the degree to which parameter values affect the accuracy of detection. Another step will be to examine if there is a need for parameter estimation based on noise level in data. Or simply put, can this method be applied to data that is not preprocessed.

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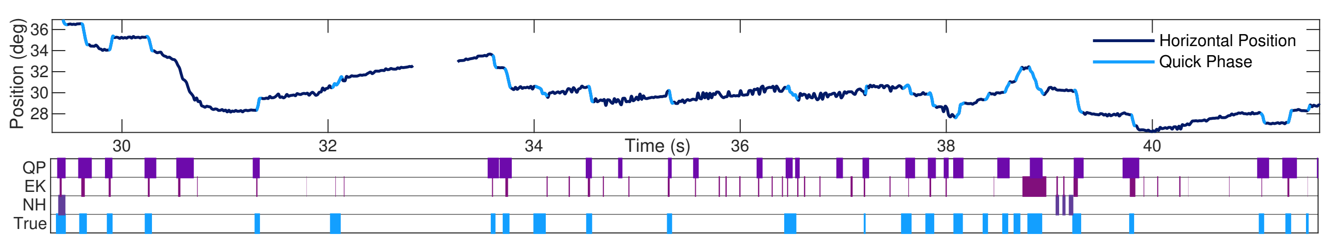

Figure 3: Proposed Quick-Phase Detection method (QP), EK and NH algorithms with indicated true labels of the quick phases.

REFERENCES

Pierre M. Daye and Lance M. Optican. 2014. Saccade detection using a particle filter. *Journal of Neuroscience Methods* 235 (Sept. 2014), 157–168. https://doi.org/10.1016/j.jneumeth.2014.06.020

M. R. Dix and C. S. Hallpike. 1952. LXXVIII The Pathology, Symptomatology and Diagnosis of Certain Common Disorders of the Vestibular System. *Annals of Otology, Rhinology & Laryngology* 61, 4 (Dec. 1952), 987–1016. https://doi.org/10.1177/000348945206100403

Ralf Engbert and Reinhold Kliegl. 2003. Microsaccades uncover the orientation of covert attention. *Vision Research* 43, 9 (April 2003), 1035–1045. https://doi.org/10.1016/S0042-6989(03)00084-1

Casper J. Erkelens and Olaf B. Sloot. 1995. Initial directions and landing positions of binocular saccades. *Vision Research* 35, 23-24 (Dec. 1995), 3297–3303. https://doi.org/10.1016/0042-6989(95)00077-R

K Hanley, T O'Dowd, and N Considine. 2001. A systematic review of vertigo in primary care. *The British Journal of General Practice* 51, 469 (Aug. 2001), 666–671. https://www.ncbi.nlm.nih.gov/pmc/articles/PMC1314080/

Roy S. Hessels, Diederick C. Niehorster, Chantal Kemner, and Ignace T. C. Hooge. 2017. Noise-robust fixation detection in eye movement data: Identification by two-means clustering (I2MC). *Behavior Research Methods* 49, 5 (Oct. 2017), 1802–1823. https://doi.org/10.3758/s13428-016-0822-1

Sabrina Hoppe and Andreas Bulling. 2016. End-to-End Eye Movement Detection Using Convolutional Neural Networks. *arXiv:1609.02452 [cs]* (Sept. 2016). http://arxiv.org/abs/1609.02452 arXiv: 1609.02452.

Athanasios Katsarkas. 1999. Benign Paroxysmal Positional Vertigo (BPPV): Idiopathic Versus Post-traumatic. *Acta Oto-Laryngologica* 119, 7 (Jan. 1999), 745–749. https://doi.org/10.1080/00016489950180360

Kattah Jorge C., Talkad Arun V., Wang David Z., Hsieh Yu-Hsiang, and Newman-Toker David E. 2009. HINTS to Diagnose Stroke in the Acute Vestibular Syndrome. *Stroke* 40, 11 (Nov. 2009), 3504–3510. https://doi.org/10.1161/STROKEAHA.109.551234

Oleg V. Komogortsev and Alex Karpov. 2013. Automated classification and scoring of smooth pursuit eye movements in the presence of fixations and saccades. *Behavior Research Methods* 45, 1 (March 2013), 203–215. https://doi.org/10.3758/s13428-012-0234-9

Alexandra I. Korda, Pantelis A. Asvestas, George K. Matsopoulos, Errikos M. Ventouras, and Nikolaos Smyrnis. 2018. Automatic identification of eye movements using the largest lyapunov exponent. *Biomedical Signal Processing and Control* 41 (March 2018), 10–20. https://doi.org/10.1016/j.bspc.2017.11.004

Andra Mihali, Bas van Opheusden, and Wei Ji Ma. 2017. Bayesian microsaccade detection. *Journal of Vision* 17, 1 (Jan. 2017), 13–13. https://doi.org/10.1167/17.1.13

Matthew S. Mould, David H. Foster, Kinjiro Amano, and John P. Oakley. 2012. A simple nonparametric method for classifying eye fixations. *Vision Research* 57 (March 2012), 18–25. https://doi.org/10.1016/j.visres.2011.12.006

Jeffrey B. Mulligan. 2018. Statistical Identification of Fixations in Noisy Eye Movement Data. *Electronic Imaging* 2018, 14 (Jan. 2018), 1–7. https://doi.org/10.2352/ISSN.2470-1173.2018.14.HVEI-528

David E. Newman-Toker, Yu-Hsiang Hsieh, Carlos A. Camargo, Andrea J. Pelletier, Gregary T. Butchy, and Jonathan A. Edlow. 2008. Spectrum of dizziness visits to US emergency departments: cross-sectional analysis from a nationally representative sample. *Mayo Clinic Proceedings* 83, 7 (July 2008), 765–775. https://doi.org/10.4065/83.7.765

Marcus NystrÃűm and Kenneth Holmqvist. 2010. An adaptive algorithm for fixation, saccade, and glissade detection in eyetracking data. *Behavior Research Methods* 42, 1 (Feb. 2010), 188–204. https://doi.org/10.3758/BRM.42.1.188

J. Otero-Millan, J. L. A. Castro, S. L. Macknik, and S. Martinez-Conde. 2014. Unsupervised clustering method to detect microsaccades. *Journal of Vision* 14, 2 (Feb. 2014), 18–18. https://doi.org/10.1167/14.2.18

Tomasz Pander, Robert CzabaÅĎski, Tomasz PrzybyÅĆa, and Dorota Pojda-Wilczek. 2014. An automatic saccadic eye movement detection in an optokinetic nystagmus signal. *Biomedical Engineering / Biomedizinische Technik* 59, 6 (Jan. 2014). https://doi.org/10.1515/bmt-2013-0137

Alex Rodriguez and Alessandro Laio. 2014. Clustering by fast search and find of density peaks. *Science* 344, 6191 (June 2014), 1492–1496. https://doi.org/10.1126/science.1242072

Thiago Santini, Wolfgang Fuhl, Thomas KÃijbler, and Enkelejda Kasneci. 2015. Bayesian Identification of Fixations, Saccades, and Smooth Pursuits. *arXiv:1511.07732 [cs]* (Nov. 2015). http://arxiv.org/abs/1511.07732 arXiv: 1511.07732.

Denis Sheynikhovich, Marcia BÃľcu, Changmin Wu, and Angelo Arleo. 2018. Unsupervised detection of microsaccades in a high-noise regime. *Journal of Vision* 18, 6 (June 2018), 19–19. https://doi.org/10.1167/18.6.19

Miika Toivanen, Kati Pettersson, and Kristian Lukander. 2015. A probabilistic real-time algorithm for detecting blinks, saccades, and ïňĄxations from EOG data. (2015), 15.

M von Brevern, A Radtke, F Lezius, M Feldmann, T Ziese, T Lempert, and H Neuhauser. 2006. Epidemiology of benign paroxysmal positional vertigo: a population based study. *Journal of Neurology, Neurosurgery & Psychiatry* 78, 7 (Dec. 2006), 710–715. https://doi.org/10.1136/jnnp.2006.100420

Raimondas Zemblys, Diederick C. Niehorster, Oleg Komogortsev, and Kenneth Holmqvist. 2018. Using machine learning to detect events in eye-tracking data. *Behavior Research Methods* 50, 1 (Feb. 2018), 160–181. https://doi.org/10.3758/s13428-017-0860-3